Chapter 1 Teacher Key — Combined

Answer Key — Chapter 1 Quiz (v1.1)

- **Source alignment:** Reader Ch.1 pp. 1–6; TG L1–L2.
- **Format:** 9 items · **16 points** · record **raw score + percentage** on the SCORE line (`___ / 16` `___ %`).
- **Slot:** Friday · Session 3 · minutes 12–25 (see the Chapter 1 pacing plan). ~13 minutes.
- **Note policy — TEACHER CHOICE: OPEN NOTE OR CLOSED NOTE.** No project-wide rule exists. Circle one on each copy before printing or announce it; do not change the policy mid-class. Open note = student's own completed Guided Notes only (no slides, no partner, no Reader).

Key (item-by-item)

Item	Pts	Correct / expected response	Acceptable alternatives	Reasoning (source)	Common misconception	What NOT to penalize
1	1	**c)** cause change	—	Energy is the ability to cause change (Reader p. 1)	energy as a substance / always visible	—
2	1	**b)** evidence	—	A change is evidence that energy is present (Reader p. 2)	"energy can be seen"	—
3	2	(1) the car changed from still to moving / it rolled · (2) **motion**	"it moved" · "motion energy"	1 pt change, 1 pt form (Reader pp. 1–3)	naming the pusher as the form	spelling, wording
4	1	**d)** motion, sound, light, heat, electrical energy	—	Five forms (Reader p. 3)	—	—

Item	Pts	Correct / expected response	Acceptable alternatives	Reasoning (source)	Common misconception	What NOT to penalize
5	2	ENERGY → **change** → **evidence**	—	Notes model N1 (ADAPTED from Reader p. 2 — the chain itself is a notes scaffold); **1 pt per blank, in order** (swapped words = 0 for those blanks)	swapping change/evidence	arrow direction wording
6	2	**Yes** (1 pt) — it has **stored energy** that can cause a change **later**, e.g. when the flashlight is switched on (1 pt)	"yes, stored energy" + any later change	Reader p. 4 (battery)	only moving objects have energy	informal phrasing
7	1	**a)** has stored energy	—	A raised object has stored energy because of its position (Reader p. 5)	stored energy = moving	—
8	3	**Yes** (1 pt) · names the observed **change** (off → on / dark → bright) (1 pt) · links the change as **evidence** of energy (1 pt)	any wording that does all three; student must use the word "change"	Reader p. 2	"I saw the energy" (redirect: you saw the change)	spelling; "proof" for "evidence" (accept, restate "evidence")
9	3	a) **YES** · b) **NO** · c) **YES** (1 pt each)	—	a) “The flag waved when the wind blew” and c) “The music got louder” are observed	treating any observation as evidence	—

Item	Pts	Correct / expected response	Acceptable alternatives	Reasoning (source)	Common misconception	What NOT to penalize
				changes (motion, sound) = evidence; b) "The desk is brown" says how it looks, not a change (Reader p. 2)		

Answer letters (MC only): Q1 c · Q2 b · Q4 d · Q7 a. Point values per item sum to **16** (1+1+2+1+2+2+1+3+3).

Scoring

- Tally raw points → write `score / 16` and the percentage (score ÷ 16 × 100, rounded to a whole percent) on the SCORE line.
- Partial credit is allowed on items 3, 5, 6, 8, 9 as listed. Do not subtract for spelling or handwriting.

Out-of-scope confirmation

- No kinetic/potential/mechanical/radiant/compression-wave/inertia.
- No transfer-vs-transformation classification, no collisions, no engineering design.

Answer Key — Chapter 1 Practice (v1.1)

- **Source alignment:** Reader Ch.1 pp. 1–6; TG L1–L2. Light switch, toy car, fan, and stored-energy rows come from the Reader or Guided Notes; the **bell** and **ice cube** rows are EXTENSION (everyday experience) — accept any reasonable observable change.

Part A — What changed?

Scenario	What changed? (accept any observable change)	Form of energy (accept any reasonable one of the five)
A light switch is flipped.	the room changed from dark to bright	light (or electrical energy)
A toy car is pushed.	the car changed from still to moving	motion
A bell is rung. (EXTENSION)	quiet → a ringing sound	sound
An ice cube sits on a warm hand. (EXTENSION)	the ice melts / gets smaller (solid → liquid)	heat

Part B — Evidence or not enough evidence?

Observation	Circle	Because … (sample)
The fan blades started spinning when we plugged it in.	EVIDENCE	we saw a change: still → moving
The battery looks shiny.	NOT ENOUGH	looking shiny is not a change that shows energy at work
The room got brighter when the light came on.	EVIDENCE	dark → bright is a change
I think the motor is strong.	NOT ENOUGH	it is an opinion; no change was observed

Part C — Stored energy

Example	Stored energy?	Change later — or happening now?
a battery in a flashlight that is switched off	**YES**	when the flashlight is switched on, the stored energy can make the bulb light (Reader p. 4)

Example	Stored energy?	Change later — or happening now?
an apple held above the ground	**YES**	if it is let go, it falls (stored energy because of its position — Reader p. 5)
a stretched rubber band	**YES**	if it is let go, it snaps back and may make a sound
a toy car rolling across the floor	**NO** — its energy is in its motion **now** (not stored)	the change is already happening

Rationale (v1.1): the v1.0 "book lying flat on a table" row was removed — whether a book on a table has stored energy depends on the reference level and is outside the Reader's scope. The toy-car row gives a clear contrast: moving now vs stored for later.

Part D — Explain (sample)

"I flipped a switch and the room changed from dark to bright. A change is evidence of energy, so the light shows energy was at work."

- Accept any response that (a) names an observed change and (b) says the change is evidence of energy. Do not require "stored energy." Accept "proof" once; restate "evidence."

Teacher Look-Fors — Energy Detectives (OPTIONAL · NOT GRADED)

- **Status: optional, formative, UNGRADED.** No points, no score line. It is never part of the required three-session plan and never required for the quiz. Use it on a review day or Friday FLEX **after** the quiz, or as an enrichment choice.
- **Source alignment:** AP 1.1 (Energy Scavenger Hunt); Reader Ch.1 pp. 1–6; TG L1.

Look-fors (teacher check, not scored)

Part	Look for	Common slip	Response
A (class example)	a real before/after: dull pencil → shorter, sharper pencil,	"I saw the energy"	"You saw the change; energy is what caused

Part	Look for	Common slip	Response
	shavings, sound		it."
A — form	electricity, or motion + sound	—	accept any reasonable form
B (hunt) — 3 rows	object, a real before and after, an energy form	copying the class example	ask for a different object
B — question	any question about energy or change	—	celebrate curiosity
C (flashlight demo) — prediction	any prediction	treating a wrong prediction as a failure	never penalize a prediction
C — observation / evidence	the light came on / the bulb changed from off to on	claiming "the battery got hot" without observing it	only credit what was observed
C — explanation	the battery has stored energy; it caused a change later	only moving objects have energy	restate: "A still battery can store energy for later."

Feedback options (no points)

- ✔ / ✔+ / "try again" on the page, or a quick verbal check during circulation.